

SEHAJ SINGH

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EDUCATION

University of Maryland, College Park

College Park, MD

BS in Computer Engineering, Minor in Entrepreneurship | GPA: 3.46

Aug. 2025 – Expected May 2028

- **Coursework:** Computer Systems, Object Oriented Programming, Signals and Image Processing, Linear Algebra, Differential Equations

TECHNICAL SKILLS

Languages: C++17, Python, C, TypeScript/JavaScript, Java, SQL, MATLAB

Technologies/Frameworks: OpenGL/GLSL, Gaussian Splatting, SAM 3, MuJoCo, PyTorch, OpenCV, NumPy, SciPy, Pandas, Scikit-learn, React, Next.js, Node.js, FastAPI, REST APIs, WebSockets, WebRTC, MediaPipe, MongoDB, OpenSearch, Git, GDB, AddressSanitizer, Bash, Linux

EXPERIENCE

Paphos Oneironautics

June 2026 – Present

Software Engineer Intern

Remote

- Replaced A* with Theta* any-angle pathfinding for a cable-driven farm robot, cutting mean waypoints from **59 to 2.1** and path cost **32%**
- Took a **5.8M**-Gaussian splat renderer from **32 to 60 FPS** with view-frustum culling and a background-thread depth sort
- Built a SAM 3 based 3D instance segmentation pipeline projecting Gaussians through **367** camera poses, reaching **F1 84.2** vs **69.3** baseline
- Assigned **195 of 197** 3D fruit instances to parent trees by nearest-surface voting, validated **83/83** against held-out labels
- Found generated 3D routes were unexecutable for a rider with no winch and constrained search to the cable-height plane, cutting worst-case node expansions from **154,000 to 458**
- Root-caused a long-standing exit crash with AddressSanitizer to ECS query handles destructing after allocator teardown

AAC Inc.

June 2026 – Aug. 2026

Software Developer Intern

Tysons, VA

- Root-caused two clamping bugs in a Python telemetry engine, cutting erroneous records from **566,708 to 0** across a **5M-record** dataset
- Proved a third bug with a controlled A/B test, forcing two servers into different states and using OpenSearch std dev (**0.031 vs 4.19**) as evidence
- Built and operated a multi-server data pipeline across **3** Linux hosts, provisioning environments and writing bulk-upload automation
- Ran a parameter study across all **7** config values to isolate the bug as code-level, not configuration-level, then shipped the fix

University of Maryland, Perception and Robotics Group

Feb. 2026 – Present

Undergraduate Research Assistant

College Park, MD

- Train and export Gaussian Splatting and NeRF models on a SLURM-scheduled GPU cluster, benchmarking reconstruction quality across competing methods
- Got an Instruct-GS2GS pipeline running end to end by resolving version conflicts across the HuggingFace and CUDA stack on cluster hardware
- Implement least squares estimation and sensor calibration algorithms for high-dimensional data in real-time robotic systems

PROJECTS

Shadow Box | TypeScript, React, Node.js, WebSockets, WebRTC, MediaPipe

shadowbox.gg

- Shipped and operate a live real-time 1v1 multiplayer browser game with peer-to-peer WebRTC video
- Wrote the netcode: a WebSocket server holds authoritative match state and resolves simultaneous move commits
- Built in-browser MediaPipe pose detection scoring player gestures at framerate; own the stack through deployment

Internia | Next.js, TypeScript, FastAPI, MongoDB

- Co-built an ML research paper recommender on the Semantic Scholar API, filtering **1M+** papers across **24** domains
- Built the Next.js and TypeScript frontend with NextAuth on a FastAPI and MongoDB backend

LEADERSHIP

Project LIFT

May 2026 – Present

Executive Board, Client & Project Operations

College Park, MD

- Co-lead client operations for a **60+** student org that has delivered **\$60,000+** in software value to local small businesses
- Run discovery calls and author scope, fit, and risk decision docs determining which clients the org takes on, targeting **8** for the year